

**PREVALENCE OF HEPATITIS B VIRUS IN PREGNANT WOMEN
ATTENDING ONE ANTENATAL CARE FACILITY IN MBABANE.**



Department of Medical Laboratory Sciences

Faculty of Health Sciences

Eswatini Medical Christian University (EMCU)

A research project submitted in partial fulfillment of the requirements for the degree of Bachelor
of Science in Medical Laboratory Sciences

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ESWATINI MEDICAL CHRISTIAN UNIVERSITY (EMCU)

**FACULTY OF
HEALTH SCIENCES**

**DEPARTMENT OF
MEDICAL LABORATORY SCIENCES**

CERTIFICATION

This thesis of Zishwili Mayibongwe Cyprian 202150350 titled: "PREVALENCE OF HEPATITIS B VIRUS IN PREGNANT WOMEN ATTENDING ONE ANTENATAL CARE FACILITY IN MBABANE" submitted to the department of Medical Laboratory Sciences, Faculty of Health Sciences, Eswatini Medical Christian University in partial fulfillment of the requirements for the degree of Bachelor of Science in Medical Laboratory Sciences [BMLS] has been approved by;

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This thesis is [Accepted] / [Rejected] by the Faculty of Health Sciences

Dean, Faculty of Health Sciences

Date

DECLARATION

I declare that "**Prevalence Of Hepatitis B Virus In Pregnant Women Attending One Antenatal Care facility In Mbabane**" is my own, all the sources that I have used or quoted have been indicated and acknowledged using complete references and that this work has not been submitted before for any other degree or other institution.

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DEDICATION

This thesis is dedicated to the pregnant women of Eswatini, whose health and well-being are at the heart of this research. May the findings contribute to a healthier future for mothers and their children, free from the burden of Hepatitis B Virus.

I also dedicate this work to my family, friends and colleagues, whose unwavering love, encouragement, and understanding provided the essential strength and motivation to pursue this academic endeavor.

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I would like to honour the Eswatini Health and Human Research Review Board (EHHRB) for granting ethical approval, ensuring the integrity and ethical conduct of this research.

More appreciation is directed to the Mbabane Public Health Unit (PHU), its dedicated staff, for their cooperation and for granting permission to access the essential antenatal care and laboratory records. Your commitment to public health made this study a success.

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ABSTRACT

Background: Hepatitis B Virus (HBV) infection poses a significant global public health threat, with mother-to-child transmission (MTCT) being a primary route for establishing chronic infection, particularly in sub-Saharan Africa. In Eswatini, there is a paucity of recent, localized data on HBV prevalence among pregnant women, which hinders the effective monitoring and refinement of antenatal care (ANC) prevention strategies.

Objectives: The study aimed to determine the prevalence of HBV infection among pregnant women attending the Mbabane Public Health Unit, describe its annual trends, and identify the age cohort with the highest burden of infection.

Methodology: A retrospective cross-sectional study was conducted by reviewing antenatal records from January 2021 to December 2024. Random sampling technique was applied. A sample size of 86 participants was selected. The study received ethical clearance from Eswatini Health and Human Research Review Board (EHHRRB).

Results: The prevalence of HBV infection among the sampled participants was 3.5% (95% Confidence Interval (CI): 0.7%-9.9%). A high rate of HIV co-infection was observed, with 33.3% of HBV-positive women also living with HIV. Critically, none of the HBV-positive women had received HBV vaccination, indicating 0% vaccination coverage. The prevalence was highest in the 28-37 years age group at 2.3%, though this finding was not statistically significant across age cohorts due to the small sample size. All positive cases were detected in the third trimester. Annual prevalence fluctuated, peaking at 14.3% (95% CI: 0.4%-57.9%) in 2022 and dropping to 0% in 2024.

Conclusion and Recommendations: The findings indicate a persistent, moderate prevalence of HBV among pregnant women in Mbabane Public Health Unit, Eswatini, compounded by a high rate of HIV co-infection and an alarming absence of vaccination. To curb MTCT and work towards the WHO's 2030 elimination targets, this study conclusively recommends the urgent strengthening of ANC programmes.

Keywords: Hepatitis B Virus, Prevalence, Hepatitis B surface antigens (HBsAg) testing, Antenatal care, Mother-To-Child Transmission (MTCT), Eswatini.

ABBREVIATION AND ACRONYMS

AGL	Antigenic Loop
ANC	Antenatal Care
rcDNA	Relaxed Circular DNA
RNP	Ribonucleoprotein
VLPs	Virus Like Particles
cccDNA	Covalently Closed Circular DNA
CHB	Chronic Hepatitis B
DNA	Deoxyribonucleic Acid
PHU	Public Health Unit
HBV	Hepatitis B Virus
HBsAg	Hepatitis B Surface Antigen
HBcAg	Hepatitis B core Antigen
HBeAg	Hepatitis B e Antigen
HBIG	Hepatitis B immune globulin
HSP	Heparan Sulphate Proteoglycan
NTCP	Sodium Taurocholate Co-transporting Polypeptide
pgRNA	Pregenomic RNA
LIS	Laboratory Information System
MTCT	Mother To Child Transmission
WHO	World Health Organisation
EHHRB	Eswatini Health and Human Research Review Board
ART	Antiretroviral Therapy

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CHAPTER ONE

INTRODUCTION

1.1 Background

Hepatitis B Virus (HBV) is a Deoxyribonucleic acid (DNA) virus that specifically infect and damage the hepatocytes, leading to a hepatotropic condition. The HBV DNA is spherical, enclosed with lipid envelop and the HBV genome is circular and double stranded. The HBV genome encodes for 7 antigens which are; polymerase P, L-Hepatitis B Surface Antigen, M-Hepatitis B Surface Antigen, S-Hepatitis B Surface Antigen, Hepatitis B core Antigen (HBcAg), Hepatitis B e Antigen (HBeAg) and X antigen (Guo *et al.*, 2023). The infectious HBV matured virion is known to be the Dane Particles (Zakrzewicz & Geyer, 2022).

The Dane particle contains the surface antigens; L-Hepatitis B Surface Antigen (L-HBsAg), M-Hepatitis B Surface Antigen (M-HBsAg) and S-Hepatitis B Surface Antigen (S-HBsAg). HBV infected blood contain the Dane particle and viral like particles (VLPs) which do not contain HBV genomic DNA rather composed of Middle Hepatitis B Surface (M-HBs), Small Hepatitis B Surface (S- HBs) and Large Hepatitis B Surface L- HBs proteins. These proteins happen to contain similar termination codon. Protein L- HBs contain other 3 proteins called preS1, PreS2 and S. Protein L- HBs have binding capacity in the form of a domain that has high affinity for sodium taurocholate co-transporting polypeptide (NTCP) which facilitate hepatocyte infection. The S- HBs are the main components making up the VLPs (Tsukuda & Watashi, 2020).

There are serologic subtypes of HBV and they are divided into 4 namely adw, ayw, adr and ayr (Li, Wei & Yang, 2024). These subtypes are based on the expressed antigen epitope. For successful treatment of HBV, the reduction of HBcAg becomes the major objective in the means to enhance induction of sustainable cellular and humoral immune response (Pourkarim *et al.*, 2014).

The pathogenesis of HBV infection begins at entry of HBV into host cell via transmission modes which could be atleast one of the following ways; mother-to-child-transmission (MTCT), blood borne transmission or sexually transmitted. The HBV attach to the heparan sulphate proteoglycan (HSPG) in cell surface of hepatocytes aided by antigenic loop (AGL) in the HBsAg domain. This is followed by L Protein binding to NTCP via the preS1 region (Nassal, 2015). Once HBV

relaxed circular DNA (rcDNA) gain entry into the cell nucleus it initiates formation of covalently closed circular DNA (cccDNA) which exists in the form of minichromosome in hepatocytes (Li, Wei & Yang, 2024). cccDNA act as a template for generation of the HBV subtypes and formation of the HBV antigens through translation.

The 3.5 kb pgRNA plays a crucial role in reverse transcription and replication. According to Hu and Liu (2017), during viral replication, the HBV-DNA polymerase attach to the stem loop near the 5' end of pgRNA. This basically forms the pgRNA-polymerase ribonucleoprotein (RNP) complex which forms part of the core shell (Hu & Liu, 2017). The core shell's 3.5 kb pregenomic RNA becomes the template for synthesis of + strand DNA, thereby forming progeny rcDNA. The rcDNA enters new host after viral budding from cell nuclei and generate more cccDNA thus maintain cccDNA and double stranded linear DNA (Li, Wei & Yang, 2024).

HBV infection presents a vast array of clinical manifestations which can arise from acute HBV infection, chronic HBV infection to extrahepatic manifestations. Acute HBV infections clinically manifest through presenting the following symptoms; jaundice due to liver's inability to process bilirubin, dark urine due to increased bilirubin level in the blood, painful joints due to deposited immune complexes into the synovial cavity the immune complexes are formed by the immune system in response to the HBsAg infection, prodromal fever and loss of appetite is another symptom during acute HBV infection which is caused by the released pro-inflammatory cytokines that interact with the hypothalamus (Du *et al.*, 2017).

The symptoms of chronic HBV infection level up to present conditions such as cirrhosis which is simply the scarring of the liver, this occurs because the liver cells are damaged by inflammation and cell mediated immunity against HBsAg on the hepatocytes (Trépo, Chan & Lok, 2014). Also, hepatocellular carcinoma which is liver cancer is common in chronic HBV infection. This is because HBV DNA can integrate into hepatocytes and cause genetic instability. Furthermore, liver failure can be seen in chronic HBV infection and it is a life threatening condition that requires immediate interventions.

Lastly, extrahepatic manifestations can include the following: HBV associated polyarteritis nodosa which is the inflammation of the blood vessels also known as vasculitis, HBV associated glomerulonephritis which is caused by the deposition of HBV initiated immune complexes into the glomeruli of the kidney (Yoo *et al.*, 2015). HBV can pose a huge health risk to individuals

with compromised immune system, these individuals could be persons with underlying comorbidities, the elderly group, most vulnerable the infants and the unborn.

HBV infection is not confined to a particular group, it can be transmitted from HBsAg positive Mother to unborn more especially when there are no means executed against this potential calamity.

HBV poses unique challenges during pregnancy, this is because the maternal immune system undergoes modifications, which potentially lead to impaired immune responses against HBV thus stimulating viral activity. However, pregnancy does not worsen the course of chronic HBV for the mother unless there is an existence of an advanced liver disease, then that is when flare up of hepatitis can occur (Sirilert & Tongsong, 2021). Furthermore, HBV infection during pregnancy has been linked to an increased risk of gestational diabetes mellitus, preterm birth, and neonatal jaundice (Jiang *et al.*, 2023).

Mother-to-child transmission (MTCT) is the predominant cause of chronic HBV infection worldwide, especially in high-prevalence regions like sub-Saharan Africa. Transmission in MTCT can occur in three pathways. The first route of MTCT of HBV is intrauterine (Prenatal) transmission, wherein HBV crosses the placental barrier to infect the foetus before birth (Pan *et al.*, 2016). This pathway involves the leakage of maternal blood (particularly HBeAg-positive blood) across a compromised placental barrier or direct placental cell infection (Bai *et al.*, 2017). Secondly, another route includes the intrapartum (perinatal) transmission that occurs during labour and delivery through infant exposure to infected maternal blood and vaginal secretions. The last pathway is commonly associated with negligence during breast-feeding; it is known as postpartum (postnatal) transmission and occurs via close contact.

Several factors significantly increase the risk of MTCT. The most crucial factor is a high maternal viral load (HBV DNA level) in the mother. The high viral load in maternal blood leads to increased HBeAg circulation and HBeAg can pass the placenta. Thus the high viral load increases the risk of transmission, and most particularly in failure of infant immunoprophylaxis (Joshi & Coffin, 2020). Another factor, maternal HBeAg positivity can induce immune tolerance in the foetus, making chronic infection more likely if transmission occurs. Also, delayed HBIG (Hepatitis B immune globulin) until more than 12 hours after birth can increase the chances of HBV transmission to the infant. Lastly, incomplete Hepatitis B vaccine doses can also contribute

to MTCT of HBV within the postpartum period in cases of direct contact with infected blood (Pan *et al.*, 2016).

The above mentioned challenges and risk factors during pregnancy bring about concerns to the country's Health system in regards to the effectiveness of the existing antenatal care (ANC) programmes. According to Abdelmola, Mohammed and Ahmed, (2023) antenatal care are services provided by a skilled healthcare professional to pregnant women to ensure the best health condition for both mother and baby during pregnancy. The services include; risk identification, prevention, management of pregnancy diseases, health education and promotion (World Health Organization, 2015). It is important that all pregnant mothers and adolescents have access to proper ANC during pregnancy to prevent mother to child transmission of HBV infection and administer vaccines whenever necessary.

Lavanchy and Kane (2016) state that HBV infection stands at number 15 of diseases that causes death around the globe, meaning HBV infection is a serious health threat globally. He also mentioned that chronic HBV infection accounts for 60-80% of the caused liver cancer in Africa (Custer *et al.*, 2004). It is also estimated that the mortality and morbidity of Chronic Hepatitis B (CHB) is approximately around 20% and it contributes hugely on increased risk for lymphomas to patients with CHB.

On another note, the prevalence of HBV infection is higher in the developing countries with up to 8% while the global prevalence estimates seats at around 3.2% (Genowska *et al.*, 2023). This data is truly a caution call for a developing country like Eswatini to revisits and review the current prevalence of HBV infection within the population. With all this being said, this study intend to specifically determine the prevalence of HBV infection among pregnant women with the aim to produce updated prevalence rate among this population.

1.2 Problem statement

Hepatitis B Virus (HBV) infection has significant adverse outcomes, and infection during pregnancy can result in life-threatening complications to both mother and child. However, there is limited information on the epidemiology of HBV in Eswatini which raises concerns about the effectiveness of the existing interventions in antenatal care programmes with regards to HBV prevention and treatment measures. The paucity of such information also calls into question the country's health system standards and suggests potentially poor awareness about the risks that

comes with vertical transmission of HBV for the future population. This study will address this through determining the prevalence of HBV infection among pregnant women resulting to enhancement of the targeted prevention efforts towards transmission of Hepatitis B Virus (HBV) infection to infants.

1.3 Significance of the study

The study is important because it will determine the Prevalence of Hepatitis B Virus (HBV) infection among pregnant women attending Mbabane PHU, thus inform HBV prevention strategies which could include vaccination of pregnant women during Antenatal care. In essence, the study will also contribute to the assessment of HBV infection among pregnant women attending the facility and also align with WHO goal of eliminating HBV vertical transmission by 2030.

1.4 Research questions

1. What is the prevalence of Hepatitis B Virus (HBV) infection among pregnant women attending the Mbabane Public Health Unit?
2. How has the prevalence of HBV infection among pregnant women attending the Mbabane Public Health Unit changed annually?
3. Which age group (18–27, 28–37, or 38–47 years) exhibits the highest prevalence of HBV infection among pregnant women attending the Mbabane Public Health Unit?

1.5 Research objectives

1.5.1 Main Objectives

Determine the prevalence of HBV infection among pregnant women attending Mbabane Public Health Unit between 2021 and 2024.

1.5.2 Specific objectives

1. Determine the prevalence of HBV infection among pregnant women attending Mbabane Public Health Unit.
2. Describe the trends of HBV prevalence from 2021 to 2024.
3. Determine the age group with the highest HBV prevalence amongst age group 18-27, 28-37 and 38-47.

1.6 Scope of the study/ Delimitations

The study was conducted at the Mbabane Public Health unit (PHU), past records were used as primary source of data, and only included pregnant women who tested for HBsAg in the period 2021-2024.

1.7 Assumptions

The credibility of the study is holistically dependant on data that is assumed to be accurately recorded by qualified laboratory technologist and reflect the true value of the participant.

1.8 Conceptual and operational definitions

- a) Hepatitis B Virus - A hepatotropic DNA virus that infects liver cells (hepatocytes), causing acute or chronic liver disease.
- b) Prevalence - The proportion of a population that has a specific characteristics, disease, or condition at a given time.
- c) Hepatitis B surface antigens (HBsAg) testing - A diagnostic method to detect the presence of HBV surface antigens in blood, confirming active HBV infection.
- d) Antenatal care- Healthcare services provided to pregnant women to monitor and promote maternal and foetal health. In this study, ANC includes HBV screening and interventions to prevent MTCT.
- e) Mother-To-Child Transmission (MTCT) - The transmission of HBV from an infected mother to her child during pregnancy, childbirth, or breastfeeding.

CHAPTER TWO

LITERATURE REVIEW

2.1 Global and Regional Burden of HBV in pregnant women

Hepatitis B Virus (HBV) infection among pregnant women constitutes a major public health concern globally, with the African regions bearing a disproportionately high burden, which necessitate a focused attention on such a matter (World Health Organization, 2022). The ongoing issue of HBV vertical transmission during pregnancy, which leads to Chronic Hepatitis B (CHB) in newborns, points out the critical need to understand HBV prevalence among pregnant women (World Health Organization, 2022). Fortunately enough, a comprehensive meta-analysis estimated that a pooled HBV prevalence among pregnant women across Africa is at approximately 5.89% (Wondmeneh & Mekonnen, 2024). However, these findings highlighted a considerable regional heterogeneity across the continent, which obviously signal the need for local data rather than relying solely on continental averages (Bai *et al.*, 2017). Even though, these broad studies effectively established the problem's scale and reinforce recommendations for universal screening and prevention strategies (World Health Organization, 2022), there is still a lack of granularity required for targeted local health planning, rendering an unrecognised gap in facility-specific knowledge. This literature review will synthesize the current knowledge to establish the context for meeting the study's objectives of determining the HBV prevalence, describe the trends of HBV prevalence (2021-2024), and state the age group with highest HBV prevalence among pregnant women at the Public Health Unit (PHU) Mbabane, Eswatini.

2.2 Continental Prevalence Estimates and their limitations

There are studies from East Africa, specifically in Ethiopia, that reported a prevalence rates which is around 5.78% in Addis Ababa (Asgedom *et al.*, 2024) and 13 % in Debre Tabor (Assefa, Kiros & Delelegn, 2023). In these studies, one study typically used a cross-sectional study designs in antenatal care (ANC) setting, and the other study utilized a systematic review and meta-analysis using biomedical databases. Although these Ethiopian studies offer useful regional benchmarks and demonstrate the common methodologies, their findings are more geographically specific and they indicate the current existing HBV prevalence in that region. Then moving closer to the Southern African part where Eswatini is located, it is evident that in

the Southern African region there is significant inconsistencies in reported prevalence, which puts more emphasis on the unreliability of this extrapolating data. For instance, there is a research paper that reported a prevalence of 5.8% in Mozambique (Mathew *et al.*, 2023), and also another study in South Africa, found an overall prevalence of 11.24% (Moonsamy *et al.*, 2025). While these 2 studies provided a geographically relevant data points, there is an existing stark differences in this data reported. The difference shows a huge regional variability, which could be potentially influenced by factors like the study setting and the differences in the population characteristics involved. This documented variation strongly indicates the gap concerning the actual and current prevalence within each specific facility across the regional countries including Eswatini.

2.3 Trends and Regional heterogeneity

Beyond the observed prevalence estimates, it is so vital to understand trends. It is because trends can be used to monitor the epidemic and the ANC intervention effectiveness across periods of time. One research study in South Western Ethiopia found an HBV prevalence of 6.0% trend over a period of five years which really reflected the good effectiveness of interventions implemented against HBV in that country (Bitew, Mekonnen & Hailu, 2025). In contrast, another longer-term study which spanned from 2009 to 2016 in Cameroon observed a significantly high HBV prevalence of 11.8% over 8 year-period, this was a call into action within their healthcare system towards their strategic interventions against viral hepatitis in ANC (Shimakawa *et al.*, 2022). Even though these studies represented the Central and North East Africa, it does not cover the Southern part of Africa, which necessitate the need to address the specific and recent trends in the region, particularly on the 2021-2024 period which is much relevant to the proposed study. With all this data and critiques, there is a visible gap in knowledge regarding recent HBV prevalence and other HBV dynamics within the Eswatini Health System.

2.4 Age Cohort and HBV Prevalence

Furthermore, some existing literature suggests that age is indeed a factor to consider when assessing HBV prevalence among younger pregnant women. A research study from Ethiopia reported that the highest seroprevalence of HBV is within the 25-34 year age group (7.13%) (Belete, Fekadie & Amere, 2024). Another study in Southern Ethiopia noted the highest rates (5%) in a 60-69 year age cohort (Juhar *et al.*, 2018). Although for both studies the statistical

significance varied, the findings supported the idea of investigating age. However, the specific peak age groups differ across both studies. It is important to note that the applicability of these findings from settings like Ethiopia to Eswatini may be limited. Most significantly, these studies did not recognise the importance of analyzing the data according to the specific age categories (18-27, 28-37, 38-47 years) selected for the proposed investigation in the Public Health Unit in Mbabane, which also represent another identifiable key knowledge gap.

2.5 Methodological approaches in ANC-Based Studies

Methodologically, another cross-sectional studies analysing data from ANC attendees, as conducted in Nigeria (Machar & Jara, 2021) and Ghana (Awiah *et al.*, 2024), represented a standard approach (in a cross-sectional manner) for estimating prevalence in this population. These studies consistently reinforce the public health rationale for robust screening and prevention programs in ANC, including timely birth dose vaccination (Awiah *et al.*, 2024). The proposed study's retrospective cross-sectional design, which will be utilizing existing laboratory records from the PHU facility, aligns well with these established epidemiological methods. However, the existing studies lack geographical relevance to Eswatini more specifically PHU Mbabane and do not provide the necessary current data covering the 2021-2024 period. Therefore, applying these proven methods to the local context is necessary to generate relevant evidence.

In summary, the current body of literature establishes HBV in pregnant African women as a high-priority public health issue with variable prevalence across the continent and within regions. Standard methodologies exist for assessing prevalence and trends, and age appears to be a relevant factor. However, significant and critical gaps remain, specifically concerning the lack of current (2021-2024), localized HBV prevalence data for pregnant women of Eswatini.

CHAPTER THREE

METHODOLOGY

3.1 Research design

This study employed a retrospective, cross-sectional study design. The study used past medical records from 2021-2024. Data was collected at a specific point in time which is a feature of a cross-sectional study.

3.2 Study setting

The study was conducted at Mbabane Public Health Unit (PHU) where antenatal care services for pregnant women are provided.

3.3 Study population and sample size

3.3.1 Target and accessible population

The study's target population was pregnant women attending Mbabane Public Health Unit for antenatal care services.

3.3.2 Sampling techniques

The study used a simple random sampling technique. SPSS random number generator was used to select the sample patient IDs from cleaned sample frame. This technique reduced selection bias and potential sampling errors thus increasing the study's reliability.

3.3.3 Sample description and sample size

The sample for the study included pregnant women who were serviced at Mbabane Public Health Unit for their antenatal care programme in the period 2021-2024. Due to time constraints, a representative sample was calculated using Cochran's formula rather than reviewing the total population of the 4-year period. The previous prevalence used in the formulae is 5.89% which was the reported pooled prevalence of 91 studies from 28 African countries (Wondmeneh & Mekonnen, 2024). The rationale behind utilization of this pooled prevalence lies in the fact that it is the latest reported pooled prevalence across Africa as a whole, thus it can be generalized to the entire continent.

Sample size: it was calculated thus:

$$\frac{Z^2 * p * (1-p)}{E^2} = n$$

$$E^2$$

Where in:

n- Required sample size

Z- Standard deviation which is the 95% confidence level.

p- Previous prevalence of 5.89%

E- Margin of error will be 5%.

Here is the complete calculation:

$$\frac{1.96^2 * 0.0589 * (1-0.0589)}{0.05^2} = \frac{0.2128}{0.0025} = 85.12$$

$$0.05^2$$

$$0.0025$$

Since we are dealing with discrete samples we are compelled to round up to the nearest next whole number rendering the sample size to be **86** participants.

3.3.4 Inclusion and exclusion criteria

Inclusion criteria: Records of consenting pregnant women >18 years that attended PHU during 2021 to 2024 and tested for HBsAg during this period.

Exclusion criteria: The study did not use incomplete records, specifically with missing HBsAg laboratory results, age and ANC year. The study does not include data with clauses that state data should not be used for research purposes.

3.4 Data collection

Data that pertains to pregnant women who attended antenatal care services at the Mbabane Public Health Unit between January 1, 2021, and December 31, 2024 was extracted from the Health Management Information System (HMIS) and this data included the following: Participant's Laboratory ID, Age, and Gestational week, HIV Status, HBV Vaccination, and Hepatitis B surface antigen (HBsAg) test results.

3.5 Data analysis

The study used Microsoft Excel for data analysis and descriptive statistics including percentages are used to describe existing trends within the data. 95% Confidence Interval (CI) for prevalence rates were calculated using the exact binomial method Clopper-Pearson (online calculator) to account for the small sample size and low number of positives cases. Tables, line and bar graphs were used in the representation of the findings of the study.

3.6 Data management

Participant's data was securely stored in a password protected personal computer, only authorized personnel had access to it. The data was coded with unique identifiers to enhance participants' privacy and the results are not traceable to any personal identification of participants.

3.7 Validity and Reliability

Validity and reliability was maintained through the use of authorized hospital data as a primary source of data. Data has been recorded in the data collection sheet as is without any alterations. The study featured a simple random sampling method to select participants.

3.8 Bias

Selection bias was minimized through the use of a simple random sampling method with the assistance of SPSS random number generator executed on patient IDs.

3.9 Ethical consideration

Ethical clearance was granted by the Eswatini Medical Christian University committee and the Eswatini Health and Human Research Review Board (EHHRRB), which included a waiver of individual informed consent. The waiver was justified as the study was a retrospective analysis of pre-existing, anonymized data from HMIS, and obtaining individual consent was impracticable given the study design. An approval letter was also acquired from Mbabane Public Health Unit permitting the authorization to access data in the facility.

3.9.1 Confidentiality

Confidentiality has been maintained by allowing data to only authorized personnel and researcher, and information was reported in a way that participants are not identifiable.

3.9.2 Anonymity

The study ensured anonymity by using unique identifiers that were assigned to each participant.

3.9.3 Privacy

Participant's data has been securely guarded in a password protected personal computer throughout the study. The password has been only been accessible to the researcher and the supervisor.

3.9.4 Beneficence

Lack of proper information directed to antenatal healthcare can lead to continuous usage of ineffective interventions within antenatal healthcare programs. The study's findings will be used to inform HBV vaccination policies thus the benefits outweigh the risks that were imposed by the study.

3.9.5 Justice

Fairness is guaranteed in the study, no particular group was unfairly excluded from research opportunity nor subjected to research risks (Emanuel, Wendler & Grady, 2021).

3.10 Study limitations

The study focused on one facility of many in the country and utilized a smaller time frame, which disqualifies capabilities to generalize the findings to the entire population. The retrospective approach of the study inhibits the findings' ability to present the current and predict the future status of population.

3.11 Dissemination of findings

The research findings will be made available to the Eswatini Medical Christian University library. Another copy will be printed and shared with the ministry of health.

CHAPTER FOUR

RESULTS

Table 1: Extract of data collected for the HBV tested pregnant women.

Patient ID	Age	Age Group	ANC Year	Gravida	Gestation Week	Most Recent HIV Result	On ART	Tested Hepatitis	Hepatitis_Result	HBV Vaccination
001	25	18-27	2022	2	40	Positive	Yes	Yes	Negative	No
002	29	28-37	2023	2	35	Negative	No	Yes	Negative	No
003	27	18-27	2022	4	29	Positive	Yes	Yes	Negative	No
004	33	28-37	2024	2	30	Positive	Yes	Yes	Negative	No
005	22	18-27	2021	2	38	Negative	No	Yes	Negative	No
006	24	18-27	2023	1	35	Negative	No	Yes	Negative	No
007	20	18-27	2023	1	34	Negative	No	Yes	Negative	No
008	23	18-27	2023	1	35	Negative	No	Yes	Negative	No
017	33	28-37	2022	5	22	Positive	Yes	Yes	Negative	Yes
022	32	28-37	2021	1	40	Negative	No	Yes	Positive	No

86 sampled participants were part of the study. The data was recorded in the data collection sheet with the fields; Patient ID, Age, Age Group, ANC Year, Gravida, Gestation Week, Most Recent HIV results, On ART, Tested Hepatitis, Hepatitis Results and HBV Vaccination Status.

4.1 Overall Prevalence of HBV

Table 2: Prevalence of HBV

Hepatitis Results						
		Frequency	Percent	Valid Percent	Cumulative Percent	95% Confidence Interval (CI)
Valid	Negative	83	96.5	96.5	96.5	(90.1%-99.3%)
	Positive	3	3.5	3.5	100.0	(0.7%-9.9%)
	Total	86	100.0	100.0		

The table above shows a total of 86 participants who were enrolled in the study. There were 3 (3.5%) participants who tested positive for HBV and 83 (96.5%) participants tested negative for HBV. The overall prevalence of HBV is 3.5% (95% CI: 0.7%-9.9%).

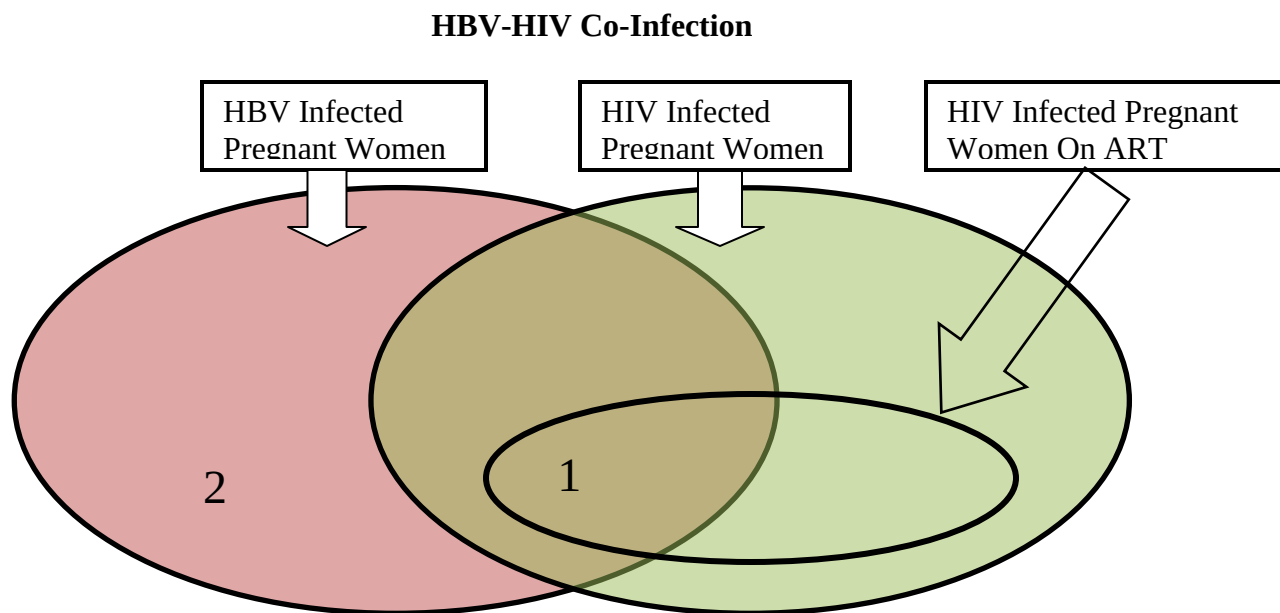


Figure 1: HBV-HIV Co-Infection

Figure 1 shows that during the study among 2/3 (66.7%) pregnant women were infected with HBV only, and among 1/3 (33.3%) of HBV positive pregnant women also had HIV and they were on ART. This yield a co-infection rate of 33.3% (95% CI: 0.8%-90.6%).

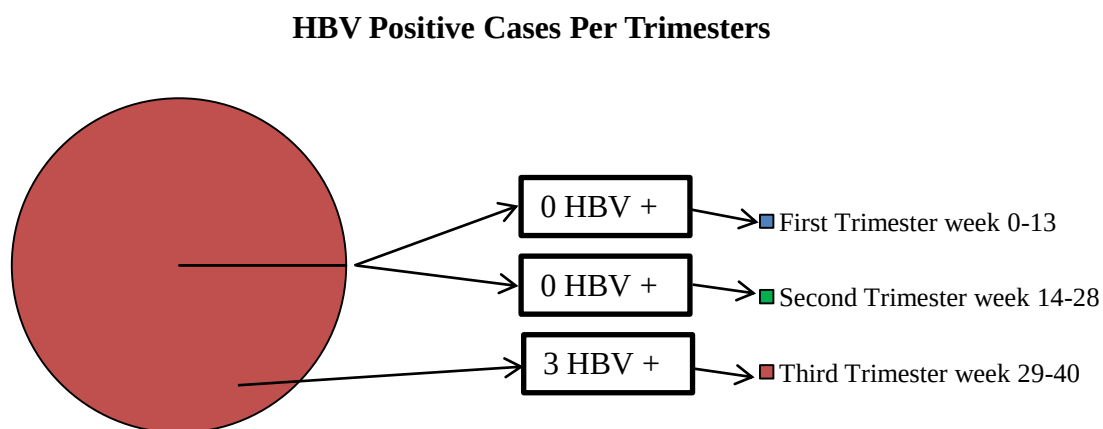


Figure 2: HBV Positive Cases Per Trimesters

The figure above highlights 3 HBV positive cases within the 3rd trimester week 29-40 period and 0 HBV positive cases in both the 1st and 2nd trimester. Most pregnant women tested HBsAg positive in the last trimester of their pregnancies.

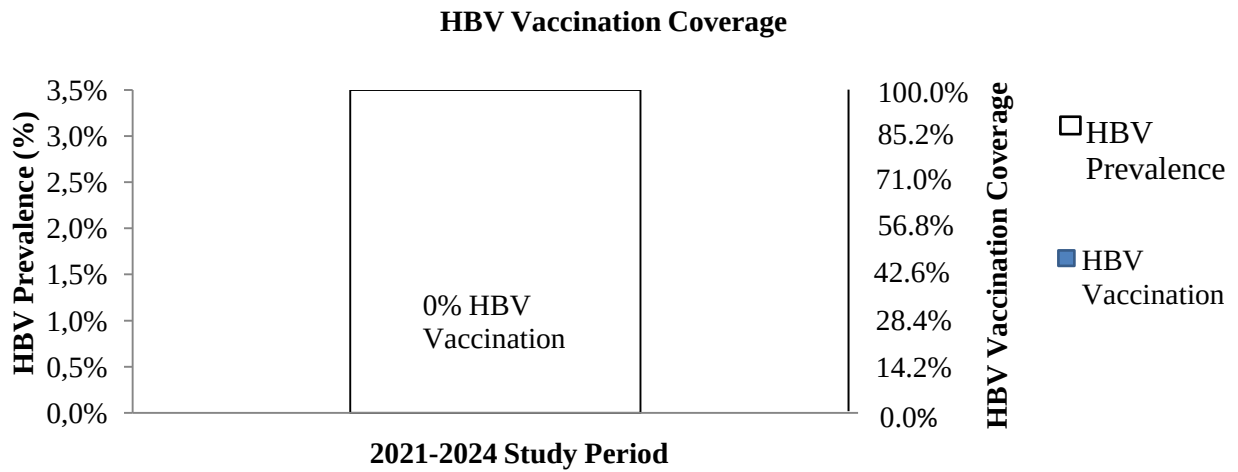


Figure 3: HBV Vaccination Coverage

The bar graph above is showing the 0% HBV vaccination coverage. The study was conducted for a period of 4 years. 86 participants were included in the study, 3 of the participants were HBV positive (3.5%) and all 3 (100%) were not vaccinated for HBV.

4.2 Annual Prevalence Trends (2021-2024)

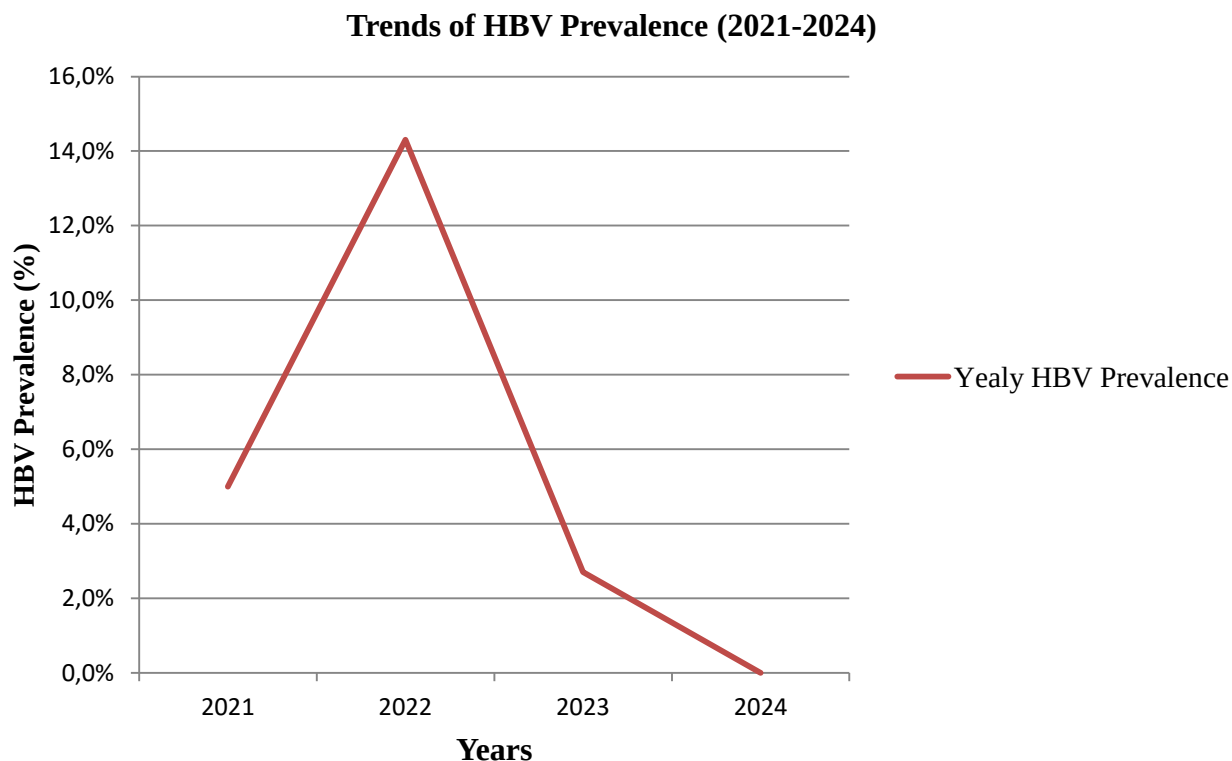


Figure 4: Trends of HBV Prevalence (2021-2024)

Figure 4 present the fluctuating annual trends for HBV prevalence over the 4 year-study-period from 2021 to 2024. The annual prevalence increased from 5% (1/20) in 2021, the prevalence reached its peak in the year 2022 with 14.3% (95% CI: 0.4%-57.9%) based on one positive case out of seven test conducted in 2022 (1/7), then a sharp decline to 2.7% (1/37) in 2023 was observed. Lastly, 2024 had a 0.00% HBV prevalence during the study.

Table 3: Annual HBV Prevalence Trends

Year	Negative	Positive	Total Tests	Annual HBV Prevalence Trend (%)	95% Confidence Interval (CI)
2021	19	1	20	5	(0.1%-24.9%)
2022	6	1	7	14.3	(0.4%-57.9%)
2023	36	1	37	2.7	(0.1%-14.2%)
2024	22	0	22	0	(0.0%-15.4%)
Total	83	3	86		

Table 3 provides more details pertaining the number of positive cases and the total tests done in each year.

4.3 Prevalence by age cohort

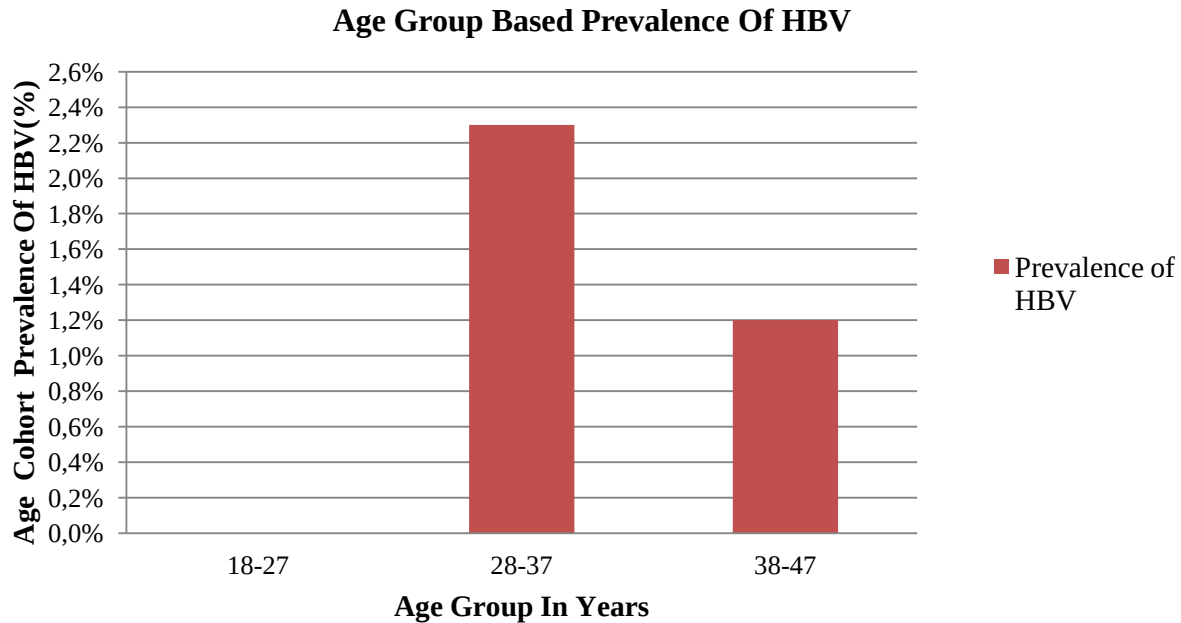


Figure 5: Age Group Based Prevalence Of HBV

Figure 5 highlights the distribution of HBV prevalence in the different age cohorts. The 28-37 age group exhibit the highest prevalence of 2.3% (95% CI: 0.1%-12.3%), followed by 38-47 years with a prevalence of 1.2%, ending with the 18-27 age group showing HBV prevalence of 0.0% (95% CI: 0.0%-4.2%).

CHAPTER FIVE

DISCUSSION AND CONCLUSION

5.1 Discussion

The overall HBV prevalence among pregnant women serviced at Mbabane Public Unit for antenatal care between 2021-2024 is 3.5% (95% CI: 0.7%-9.9%). There were 86 participants in the study, 3 of the participants tested HBV positive and 83 participants were HBV negative. The wide confidence intervals around the prevalence estimate indicate the imprecision inherent in the study due to the smaller sample size and low number of positive cases. Therefore, the point estimate of 3.5% should be interpreted with caution, as the true prevalence in the broader population can be as low as 0.7% or as high as 9.9%. The prevalence of 3.5% is similar to that of 3.5% reported for Chronic HBV infection among women of reproductive age globally (Mustapha *et al.*, 2020), however, the direct comparability is limited. As the upper bound of the CI 9.9% exceeds the pooled prevalence of 5.89% reported by Wondmeneh & Mekonnen (2024) and the lower bound is very lower.

The study found that amongst the 3 HBV positive participants, 1 was co-infected with HIV. The study deduced that around 33.3% of HBV infected pregnant women at Mbabane Public Health Unit were co-infected with HIV. The HBV-HIV co-infection rate reported by this study is within the 19-40% co-infection rate range reported by different studies in Ethiopia and is a very high rate (Tesfu, Belay & Habtemariam, 2022). This critically high HBV-HIV co-infection rate of 33.3%, despite its wide confidence interval (0.8%-90.6%), highlights a potential public health crisis that demands immediate investigation with a larger study. According to the WHO consolidated guidelines on hepatitis B and HIV, classification of HBV is in the following manner; <5% co-infection rate is low burden requiring routine screening, 5-9.9% is moderate burden requiring enhanced monitoring, and = or >10% is considered high burden and requires universal HBV screening and mandatory HBV vaccination for all pregnant women (World Health Organization, 2022).

The study went further to investigate the distribution of HBV positive cases within the trimesters of the pregnant women. According to the findings of the study, the third trimester week 29-40 had the highest HBV positive cases. All the 3 (100%) HBV positive participants of the study

were tested and found HBV positive during their third trimester week 29-40. Another study conducted in the University of Hong Kong, China, suggested that the highest HBV positive cases are recorded within the second and third trimester (Lao, 2020). These findings foster the importance of early diagnosis which is achievable through early screening protocols and HBV vaccination programmes, which could be implemented in the first trimester week 0-13 for antenatal care services.

The study also assessed HBV vaccination amongst pregnant women at Mbabane Public Health unit. A concerning 0% HBV vaccination coverage amongst the pregnant women at Mbabane Public Health Unit is reported by the findings of the study. Among the 3 HBV positive participants none of them received HBV vaccination. The absence of vaccination directly contributes to the perpetuation of HBV infection, including vertical transmission (Belyhun *et al.*, 2016). The 0% coverage as reported by the study findings is a simple risk maximized and a huge drawback towards the goal of eliminating transmission of HBV transmission by 2030. However, the 0% documented vaccination rate possibly suggests a limitation in the completeness and retrievability of historical vaccination records within the facility's HIMS, rather than a confirmed lack of immunization in the sampled women.

The study have also assessed the annual HBV prevalence trends over the study period. Significant fluctuations were observed and the highest prevalence was in 2022 at 14.3% (95% CI: 0.4%-57.9%). Due to the extremely wide confidence of interval, this finding has high level of uncertainty since only 7 tests were conducted and only one was a positive case. The annual prevalence increased from 5% (1/20) in 2021 to 14.3% (1/7) in 2022, then a sharp decline to 2.7% (1/37) in 2023 was observed, and a flat 0.00% HBV prevalence in 2024. The observed fluctuating behavior of HBV prevalence over the study period is likely influenced by the covid19 intensive screening establishments, which raised more awareness of how vital it is to undergo check-ups (Centers for Disease Control and Prevention, 2018). That is why much assessment was done during the covid19 era, and, an antagonistic effect is observed during the post covid19 era where people lose the courage to take routine check-ups (World Health Organization, 2021).

Focused assessment on which age group possesses the highest HBV prevalence amongst age group 18-27, 28-37, and 38-47 of age in years was investigated. The study revealed the cohort

28-37 of years as having the highest HBV prevalence 2.3% (95% 0.1%-12.3%) and this aligns with global data on HBV epidemiology. This cohort 28-37 years has been often identified by many studies including this one as having a high HBV prevalence due to being in an age range of peak sexual activity and proneness to cohort effect: as this group was largely born before the widespread implementation of the universal infant HBV vaccination (Centers for Disease Control and Prevention, 2018). Studies have shown that incidence of sexually transmitted infections, including HBV, peaks in young and middle-aged adults 28-37 years (Schweitzer *et al.*, 2015). The 38-47 years age group remains the second highest age group with HBV prevalence of 1.2%. This age group has reduced HBV prevalence when compared to the age group 28-37 years. A review of sexual behaviour data indicates that the number of new sexual partners and overall sexual activity tends to decrease with advancing age, thereby reducing the risk of exposure to sexually transmitted infections such as HBV (Wang *et al.*, 2023). The analysis also revealed a non-linear relationship between age and HBsAg prevalence. The 18-27-year-old cohort exhibited the lowest prevalence of 0% (95% CI: 0.0% - 4.2%) HBV prevalence. Despite the potential upper bound limits of 0.4% which means the true prevalence could be non-zero, the 0% prevalence in the 18-27 years age group is a powerful finding and is most convincingly explained by the success of vaccination programmes. Many countries introduced universal HBV vaccination for infants in the 1990s and early 2000s. The 18-27-year-old participants, born approximately between 1996 and 2005, are likely the first generation to benefit broadly from this public health intervention, resulting to the dramatic decline in HBV prevalence among vaccinated cohorts, and reducing it to near zero in children and young adults (World Health Organization, 2016). For this reason, it is equally important to implement the universal HBV screening protocol and HBV vaccination for all pregnant women of all age groups. This practice will ensure excellent and sustainable HBV infection monitoring in this fragile population.

5.2 Limitations

Notwithstanding the contribution of this study, this study has several limitations that needs to be taken into considerations when interpreting the findings. To begin with, the retrospective cross-sectional design, reliant on completeness of pre-existing HMIS records, constrains the ability to establish causality and may have led to the underreporting of variables such as HBV vaccination status. Furthermore, the relatively small sample size (n=7 in 2022), makes the annual trend data statistically unreliable, and the peak in 2022 might be an artifact of the small sample size. This is

directly reflected in the wide 95% confidence Interval for the prevalence estimate and the 0% prevalence reported for 2024, which resulted from the sample size rather than a definitive cessation of HBV transmission.

Finally, the focus on a single antenatal facility in Mbabane Public Health Unit means the results cannot be generalized to the wider population of pregnant women of Eswatini, and this brings about the need for a multi-centre study to confirm these findings.

5.3 Conclusion

This study aimed to determine the prevalence, annual trends, and age specific burden of Hepatitis B Virus infection among pregnant women at the Mbabane Public Health Unit from 2021 to 2024. The findings confirm a measureable HBV prevalence of 3.5% within this antenatal population, with fluctuating annual trends and the highest burden identified among women aged 28-37 years.

The results indicate a persistent public health challenge, compounded by a critical absence of vaccination and a high rate of HBV-HIV co-infection. These factors collectively signify a substantial risk for mother-to-child transmission and highlight an urgent need to strengthen integrated antenatal prevention strategies, including universal screening and vaccination, to safeguard maternal and child health.

5.4 Recommendations

Given the findings of the study on pregnant women which projected the 3.5% HBV prevalence, 33.3% of HBV positives co-infected with HIV, and the 0% HBV vaccination coverage with about 5 gravida exposed to HBV infection, it is important to acknowledge the criticality of these findings to public health. Thus recommendations are deemed essential they may involve foundational strategies, and core interventions to all pregnant women and those co-infected. At Facility level implementation of foundational strategies can include universal HBV screening for all pregnant women, integration of HBsAg testing into standard antenatal blood testing panel for all pregnant women at their first visit along with HIV and syphilis testing, this will align well with the WHO's triple elimination initiative. Merge HBV services into existing and functional HIV prevention of mother to child transmission (PMTCT) infrastructure, this can aid implementation easier and effective.

With the 0% HBV vaccination coverage as indicated in the study, core interventions such as

vaccination and prevention methods are to be firmly established. Provision of antiviral prophylaxis to high-risk HBV positive pregnant women from the first trimester is a necessity to prevent MTCT. This provision can even provide dual protection to the co-infected pregnant women already on Tenofovir disoproxil fumarate TDF-based ART for HIV. As a facility it is important to ensure that infants receive the universal Hepatitis B Birth dose for infants born to infected mothers and complete the vaccine series. This will prevent transmission and development of chronic HBV infection. These are key components of the global expanded programme on immunization and vital for achieving the WHO's 2030 elimination target. If possible enough, HBV vaccination programme could also be exercised at national level to the entire population and be coupled with the ongoing Human Papilloma Virus (HPV) vaccination programme that target the younger female child, thus more females will be accessed and vaccinated for HBV too. With these recommendations and their implementation at the Mbabane Public Health Unit and to the entire National Health System elimination of HBV by 2030 can be possible.

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APPENDIX 1-RESEARCH PROJECT BUDGET

Item (Quantity)	Costs (E)
Data collection costs	
Travelling costs (20 per day for 15 days)	300
USB 64GB	350
Lunch (E45 per day for 15 days)	675
Dissemination of findings costs	
Printing	200
Binding	600
Miscellaneous costs	
Contingency fund	100
Communication (250 minutes and 2 GB data bundles)	100
Total	2,325

APPENDIX 2-STUDY TIMETABLE

ACTIVITY	FEB 2025	MAY 2025	JUN 2025	AUG 2025	OCT 2025	OCT 2025	OCT 2025
TOPIC SUBMISSIONS	7 February 2025						
PROPOSAL SUBMISSION TO DEPARTMENT		23 May 2025					
PROPOSAL DEFENCE TO DEPARTMENT		28 May 2025					
ETHICAL CLEARANCE			23 June 2025				
DATA COLLECTION				5 August 2025			
NARRATIVE REVIEW					3 October 2025		
SUBMISSION OF RESEARCH PROJECT						20 October 2025	
NARRATIVE REVIEW CORRECTIONS							

APPENDIX 3-DATA COLLECTION SHEET

[illegible]

APPENDIX 4-ETHICAL CLEARANCE APPLICATION LETTER

Zishwili Mayibongwe Cyprian
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29 May 2025

The Chairperson
Eswatini Health and Human Research Review Board (EHHRRB)
P O Box 5
Mbabane, Eswatini

Dear Sir/Madam

Application for Ethical Clearance.

I am writing to respectfully request ethical clearance from the Eswatini Health and Human Research Review Board (EHHRRB) to conduct a retrospective cross-sectional study titled: "Prevalence of hepatitis B virus among pregnant women attending one antenatal care facility in Mbabane."

This study aims to determine the prevalence of Hepatitis B virus (HBV) infection among pregnant women who attended antenatal care (ANC) services at the Mbabane Public Health Unit in the past 4 years. Understanding the burden of HBV in this population is crucial for informing local public health strategies related to antenatal care services, guiding resource allocation for prevention and control measures, and ultimately improving maternal and child health outcomes in Eswatini.

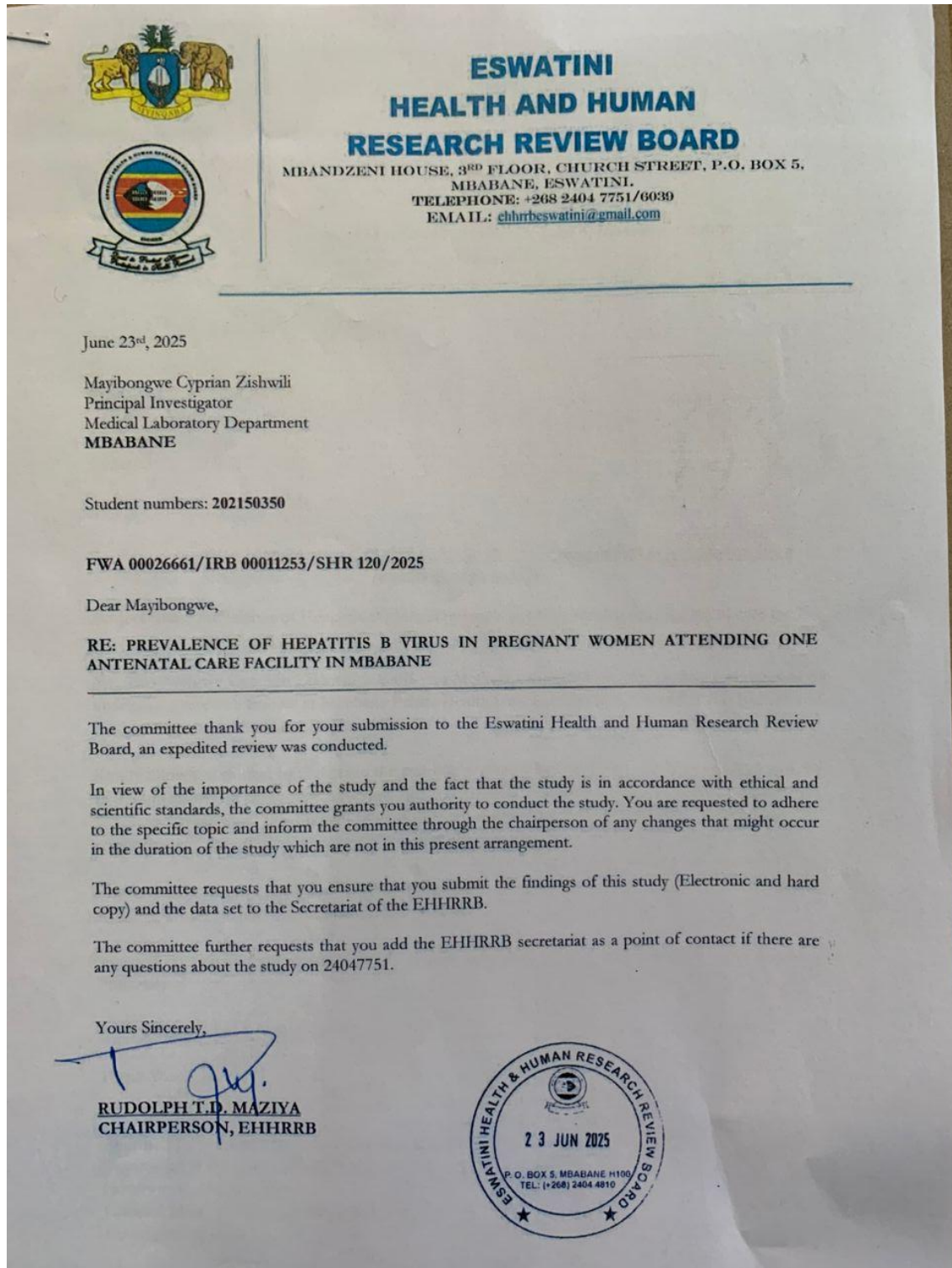
The proposed study will be a retrospective cross-sectional review of existing antenatal care records at the Mbabane Public Health Unit. Data will be collected from 1 January 2021 to 31 December 2024. The data to be extracted will include Participant laboratory ID, Age, Educational Status, Marital Status, Gestational Age, HIV Status, Blood Transfusion History, HBV Vaccination, and Hepatitis B surface antigen (HBsAg) test results. The findings of this study will be shared with the Eswatini Medical Christian University library and the Ministry of Health to contribute to evidence-based healthcare planning.

We are committed to conducting this research in accordance with the ethical principles outlined by the EHHRRB guidelines. This study will only commence upon receiving formal ethical approval from the Eswatini Health and Human Research Review Board.

We kindly request the Board to review our application. Thank you for considering our application.

Your faithfully
M.Zishwili
Zishwili Mayibongwe Cyprian
Student ID 202150350
Institution: Eswatini Medical Christian University

APPENDIX 5-ETHICAL CLEARANCE APPROVAL LETTER



APPENDIX 6-DATA COLLECTION RESEARCHER'S REQUEST LETTER

Zishwili Mayibongwe Cyprian
Eswatini Medical Christian University
P O Box A624
Swazi Plaza, Mbabane H101, Eswatini
mayibongwecy@gmail.com or +26876276487
29 May 2025

Senior Medical Officer
Mbabane Public Health Unit
P O Box 1119
Mbabane

Dear Sir/Madam

Request for Permission to Conduct a Research Study at Mbabane Public Health Unit.

I am writing to formally request permission to conduct a retrospective cross-sectional study at the Mbabane Public Health Unit. The title of the proposed research is: "Prevalence of hepatitis B virus among pregnant women attending one antenatal care facility in Mbabane."

The primary objective of this study is to determine the prevalence of Hepatitis B virus (HBV) infection among pregnant women who have received antenatal care (ANC) services at the facility.

To undertake this study, we require access to specific historical patient data spanning the period from 1 January 2021, to 31 December, 2024. The necessary data will be carefully extracted from Antenatal Care (ANC) records in HIMS, this include Age, Educational Status, Marital Status, Gestational Age, HIV Status, Blood Transfusion History, HBV Vaccination, and Hepatitis B surface antigen (HBsAg) test results.

We are fully committed to upholding the highest ethical principles throughout this research study. No data collection activities will begin until this approval is granted. The data accessed will be used exclusively for the stated research purposes. We will coordinate closely with the relevant staff to ensure that the data collection process is conducted efficiently and accurately with minimal possible disruption to the routine activities of the PHU facility

Upon completion of the research, we would be pleased to share a copy of the findings with the Mbabane Public Health Unit.

Thank you for considering our request. We look forward to your positive response and the potential for a fruitful collaboration.

Your faithfully
M.Zishwili
Zishwili Mayibongwe Cyprian
Student ID 202150350
Institution: Eswatini Medical Christian University

APPENDIX 7-DATA COLLECTION DEPARTMENTAL REQUEST LETTER



Eswatini Medical Christian University

P. O. Box A624 Swazi Plaza, Mbabane, Eswatini

E-mail: info@emcu.ac.sz

Lomkiri Portion 69 of Farm 73, Zone 4, Mbabane, Hhohho

2nd of July 2025

Our Ref: 2024/25/29-39

Senior Medical Officer
Mbabane Public Health Unit
P.O. Box 1119
Mbabane
Eswatini

Dear Sir/Madam,

RE: Requesting Mayibongwe Cyprian Zishwili (202150350) to collect research data for a project at your facility

Project Title: Prevalence of Hepatitis B virus in pregnant women attending one antenatal care facility in Mbabane

Mr. Mayibongwe Cyprian Zishwili is a 4th-year Medical Laboratory Science student and requests to undertake a research project at Mbabane Public Health Unit as a partial fulfilment of the Bachelor of Medical Laboratory Sciences Degree.

Kindly allow the student to undertake the research project. The EMCU Research Committee and the Eswatini Health Human Research Review Board (EHHRB) scrutinised and approved the proposal. We therefore assure that the student will fulfil all ethical guidelines, including maintaining the confidentiality and integrity of patients as well as the data gathered.

The data collection process will commence following your approval and is anticipated to be completed within 6 weeks, guided by your institution's policies.

Attached are supporting documentation (Ethics approval letter from EHHRB, and Research Proposal).

We hope to get the desired assistance.

Thank You

Therese Mulyangote - Head of Department (HOD)
Department of Medical Laboratory Sciences,
Faculty of Health Sciences,
Eswatini Medical Christian University (EMCU)
Email: therese.emcu@gmail.com



APPENDIX 8-DATA ACCESS AUTHORIZATION LETTER

Telegrams:
Telex:
Telephone: (+268 24043506)
Fax:



Mbabane Public Health Unit
P O BOX 1119
Mbabane

SWAZILAND

THE KINGDOM OF SWAZILAND

22nd July 2025

Mayibongwe C. Zishwili
Eswatini Medical Christian University
P.O. Box A624
Swazi Plaza Mbabane

Dear Madam

RE: AUTHORIZATION TO CARRY OUT A RESEARCH STUDY.

Mayibongwe Cyprian Zishwili who is a student at the Eswatini Medical Christian University is granted permission to conduct a study on **“Prevalence of Hepatitis B virus in pregnant woman attending one antenatal care facility in Mbabane”** at the Mbabane Public Health Unit.

Thanking you in advance.

Yours faithfully

ZANELE CINDZI
IN-SERVICE TRAINING COORDINATOR